



COURSE OBJECTIVES AND SCOPE: To develop proficiency in applying principles of thermal and fluid analysis to situations commonly found in industry. Areas of application include piping systems, heat exchangers, turbomachinery, and power cycles.

CLASS MEETING INFO:

Section	Day	Start Time	End Time	Room
Lecture	Tuesday Thursday	6:10PM	7:30PM	Pandora 146
Lab	Thursday	7:40AM	9:00PM	Pandora 116 and others

INSTRUCTOR:

T. Sean Tavares, PhD.

Associate Professor, Mechanical Engineering Technology

Department of Applied Engineering and Sciences

Pandora Rm. 103

Office Phone: (603) 641-4322

E-mail: Sean.Tavares@unh.edu (Preferred contact method. Will respond usually within 24 hours.)

OFFICE HOURS: Tuesday 4:30PM – 5:30PM, and by appointment.

REQUIRED RESOURCES: There is one required textbook:

Fundamentals Thermal-Fluid Sciences, 6th Ed. by Cengel, Cimbala, and Ghajar. ISBN 13: 9781260716979

BOOKS ON LIBRARY RESERVE: The following supplemental materials have been placed on reserve in the library. You do not need to purchase them.

1. *Introduction to Thermal and Fluids Engineering*, Kaminski & Jensen (text formerly used in ET 645)
2. *Experimental Methods for Engineers*, J.P. Holman; useful reference for labs
3. *Engines, Energy, and Entropy*, J.B. Fenn; supplemental reference for power cycles

COURSE CREDITS & FORMAT: This is 4 credit course. Classes will consist of lectures and labs. The labs involve the application of concepts taught in lecture and also some new material related to thermal and fluid sciences. The modality will be in person unless unforeseen circumstances require a change to remote format.

COURSE MANAGEMENT: The course is administered using the Canvas (myCourses) learning management system. <https://mycourses.unh.edu>



COURSE GRADE: The course grade will be determined using the following weighting of assignment groups:

Homework	32%	Tests (2 planned)	30%
Lab Projects	17%	Final Exam	15%
Practice Problems	6%		

CALCULATION OF LETTER GRADES: Letter grades for the course correspond to the following percentage points in the Canvas gradebook:

Letter Grade	Range
A	100% to 94%
A-	< 94% to 90%
B+	< 90% to 87%
B	< 87% to 84%
B-	< 84% to 80%
C+	< 80% to 77%
C	< 77% to 74%
C-	< 74% to 70%
D+	< 70% to 67%
D	< 67% to 64%
D-	< 64% to 61%
F	< 61% to 0%

ASSIGNMENTS AND COURSE SPECIFIC POLICIES: The following is a brief description of the graded assignments for the course along with some course specific policies:

Lectures: Students are responsible for all material covered in class and in reading assignments. Regular attendance at lectures and participation in discussions is strongly encouraged.

Homework: Homework assignments make up a substantial component of the course grade. Submissions will be files uploaded to Canvas (unless instructed otherwise). There will be an assignment Due Date followed by a short Grace Period. Assignments received after the Due Date but before the end of the Grace Period will be subject to a 10% penalty. No credit will be given for an assignment after the end of the Grace Period.



Labs: Labs will involve a combination of individual work and working in groups. The lab component will involve various combinations of making measurements, relating data to physical concepts, and developing analytical skills relevant to experimental work. Closed Toe Shoes and use of appropriate Personal Protective Equipment (safety glasses, etc.) are required in the lab. It is the student's responsibility to make up any missed lab work.

Practice Problems: These are short problems and conceptual questions designed to reinforce course material in a timely manner. Practice problems will be assigned, completed, and discussed during class time with rare exceptions. You can think of the Practice Problem component of the grade as points for class participation.

Tests & Final Exam: Tests will typically be given during scheduled lecture time and/or lab periods. (Part or all of one or more tests may be given in take-home format.) If a student needs to miss a test for any reason, the instructor must be notified prior to the session in which it is given. If an exam is missed without prior notification the student will receive a grade of zero. In extreme circumstances and if no prior notification is possible, the student must contact the instructor within 24 hours of the missed test or exam and submit in writing the reason that the scheduled test period was missed. See the section on Accommodations if you have circumstances that may require modified testing.

REQUIREMENTS FOR ASSIGNMENTS: For homework assignments, lab reports, practice problems/quizzes, tests, and final exam:

- (1) Show your problem-solving process. Briefly describe Define variables clearly and use sketches or graphs where appropriate.
- (2) the concepts, assumptions, and other resources you are using.
- (3) Clearly identify your final answers.
- (4) Submit your work in the format requested for the assignment.
- (5) Submit assignments as a **single coherent file** unless requested to do otherwise.
- (6) You are responsible for the **content and legibility** of all submitted assignments. Check to make sure the file you have uploaded corresponds to the given assignment and is **easily readable when printed**. Illegible content will not be considered for grading.
- (7) Tests and Final Exam are required to be **individual efforts**. No collaboration!
- (8) Collaboration on Homework, Labs, and Practice Problems is permitted.

ET MINIMUM GRADE REQUIREMENT: Starting with students entering the ET program in the fall of 2022, the following minimum grade requirement applies:

Each course required in the major must be completed with a minimum grade of C-. Students must attain a minimum GPA in the major of 2.0.

More information about University Requirements for Graduation can be found at:

<https://catalog.unh.edu/srrr/academic-policies/requirements-graduation/>



RELEVANCE TO ET PROGRAM ACCREDITATION: The UNH Engineering Technology Program is accredited by the Accreditation Board for Engineering and Technology (ABET). Learning objectives for ET courses are aligned with the ABET Outcomes for baccalaureate degree programs in Engineering Technology.

This course is designed to address the following specific MET Program Criteria for Baccalaureate Level Programs described in the section entitled *II. Program Criteria*.

- c. Perform selection, set-up, and calibration of measurement tools/instrumentation;
- d. Elements of differential and integral calculus;
- i. Thermal sciences (such as thermodynamics, fluid mechanics, heat transfer, etc.);
- l. Technical communications typically used in preparation of engineering proposals, reports, and specifications.



Tentative Agenda for Spring 2026

Semester Week	Date - Day	Tentative Agenda
Week 1	Jan 20 – T Jan 22 – R	Lecture – Course Introduction & Internal Flows, Ch. 14 Lecture – Internal Flows, Ch. 14 Lab Period – NO LAB MEETING
Week 2	Jan 27 – T Jan 29 – R	Lecture – Internal Flows, Ch. 14 Lecture – Internal Flows, Ch. 14 Lab Period – Lab Session
Week 3	Feb 3 – T Feb 5 – R	Lecture – Internal Flows, Ch. 14 Lecture – Internal Flows, Ch. 14 Lab Period – Lab Session
Week 4	Feb 10 – T Feb 12 – R	Lecture – Properties of Pure Substances, Ch. 4 Lecture – Properties of Pure Substances, Ch. 4 Lab Period – Lab Session
Week 5	Feb 17 – T Feb 19 – R	Lecture - Properties of Pure Substances, Ch. 4 Lecture - Properties of Pure Substances, Ch. 4 Lab Period – Lab Session
Week 6	Feb 24 – T Feb 26 – R	Lecture - Properties of Pure Substances, Ch. 4 Lecture - Energy Analysis of Steady Flow Devices, Ch. 6 Lab Period – Lab Session
Week 7	Mar 3 – T Mar 5 – R	Lecture - Energy Analysis of Steady Flow Devices, Ch. 6 Lecture - Energy Analysis of Steady Flow Devices, Ch. 6 Lab Period – Lab Session
Week 8	Mar 10 – T Mar 12 – R	Lecture - Energy Analysis of Steady Flow Devices, Ch. 6 Lecture & Lab Periods – TEST #1
Week 9	March 16-20	SPRING BREAK – NO CLASSES
Week 10	Mar 24 – T Mar 26 – R	Lecture - Energy Analysis of Steady Flow Devices, Ch. 6 Lecture - Energy Analysis of Steady Flow Devices, Ch. 6 Lab Period – Lab Session
Week 11	Mar 31 – T Apr 2 – R	Lecture - Energy Analysis of Steady Flow Devices, Ch. 6 Lecture – Power and Refrigeration Cycles, Ch. 9 Lab Period – Lab Session
Week 12	Apr 7 – T Apr 9 – R	Lecture – Power and Refrigeration Cycles, Ch. 9 Lecture – Power and Refrigeration Cycles, Ch. 9 Lab Period – Lab Session
Week 13	Apr 14 – T Apr 16 – R	Lecture – Power and Refrigeration Cycles, Ch. 9 Lecture & Lab Periods – TEST #2 Lab Period – Lab Session
Week 14	Apr 21 – T Apr 23 – R	Lecture – Power and Refrigeration Cycles, Ch. 9 Lecture – Power and Refrigeration Cycles, Ch. 9
Week 15	Apr 28 – T Apr 30 – R	Lecture – Power and Refrigeration Cycles, Ch. 9 Lecture – Course Wrap-Up and Review for Final
Week 16	May 5 – T May 7 – R	READING DAY – NO CLASSES Lecture Period – FINAL EXAM
Week 17	May 12 – T	Lecture Period – Contingency Day for Final



SELECTED UNIVERSITY POLICIES: The handbook of *Student Rights, Rules, and Responsibilities* is now incorporated in the Academic Catalog. See: <https://catalog.unh.edu/srrr/>
Brief descriptions of a few specific policies are given below, along with links to more complete information.

ACADEMIC INTEGRITY: Academic integrity is a core value at the University of New Hampshire. The members of its academic community both require and expect one another to conduct themselves with integrity. This means that each member will adhere to the principles and rules of the University and pursue academic work in a straightforward and truthful manner, free from deception or fraud. The policy can be found at the following location: [University Academic Integrity Policy](#)

ATTENDANCE: Class attendance is important for your learning. You are responsible for all course assignments and meeting all deadlines unless exceptions are agreed upon with the instructor ahead of time. If you need to miss a class for a planned activity, let the instructor know ahead of time. See the [Attendance and Class Requirements](#) policy in the undergraduate catalog.

CREDIT HOUR POLICY: This syllabus reflects the federal definition of a credit hour, which entails a minimum 3 hours of engaged time per week per credit over a 15-week semester. Examples of engaged time include class time, assignments, examinations, laboratories, participation in course-related experiences (attending a talk or performance, speakers and events, fieldwork, etc.), conferences, and office hours. Student work reflects intended learning outcomes and is verified through evidence of student achievement. For more information, please see: [NECHE policy on Credits and Degrees](#).

EXTENDED ABSENCES: If you are dealing with an unexpected, extenuating circumstance that will keep you out of class or affect your performance for more than a day or two, reach out to Lisa Enright, Assistant Dean of Student Success, at lisa.enright@unh.edu to request a letter be sent to all your faculty.

If you are required to miss significant class time, you will be provided temporary academic supports so that you can continue to make satisfactory progress in this course. Please contact your course instructor to discuss the specific types of supports that will be implemented during your absence.

TUTORING SERVICES: Knack is a peer-to-peer tutoring platform that is available to all enrolled students for all undergraduate courses in Durham and Manchester at no cost to students. Students looking for additional assistance outside of the classroom are advised to consider working with a peer tutor through Knack. UNH has partnered with Knack to provide students with access to verified tutors who have successfully completed your course. To view available tutors, visit unh.joinknack.com [Links to an external site.](#) and sign in with your student account. Questions about Knack Tutoring can be sent to Stephanie Kirylych, Director of Advising, at stephanie.kirylych@unh.edu.



STUDENT ACCESSIBILITY SERVICES: According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate his/her/their disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Student Accessibility Services (SAS) located on the Manchester campus in room 417 or sas.office@unh.edu.

Accommodation letters are created by SAS with the student. Please follow up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS but are under no obligation to provide retroactive accommodations.

For more information refer to www.unh.edu/sas or contact SAS at 603.862.2607, 711 (Relay NH) or sas.office@unh.edu.

CONFIDENTIALITY AND MANDATORY REPORTING OF SEXUAL VIOLENCE OR HARRASSMENT: The University of New Hampshire and its faculty are committed to assuring a safe and productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university's Title IX Coordinator (Bo Zarycky, Bo.Zarycky@unh.edu, 603-862-2930/1527 TTY). Faculty, staff or students on the Manchester campus can also contact Lisa Enright, Deputy Title IX Coordinator (lisa.enright@unh.edu; 603-641-4336) to report any incidents of sexual violence and harassment shared by students.

If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can contact the [SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy](#) at (603) 862-7233/TTY (800) 735-2964, as well as, Caroline Young, SHARRP Center Advocacy Expanded Services Coordinator for UNH Manchester (caroline.young1@unh.edu; room 417; Available in person Mondays 9 am to 4-pm and available by appointment (in person and virtually) by emailing caroline.young1@unh.edu). Individuals can also access Reach Crisis Services NH 603-668-2299 (24 hours), 77 Sundial Ave., Suite 306W, Manchester, NH.

For more information about what happens when you report, how the university treats your information once a report is made to the Title IX Coordinator, your rights and reporting options at UNH (including anonymous reporting options) please visit [student reporting options](#). [The uSafeUS app](#) is also available for students to keep reporting options and resources easily accessible on their phones.

Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu or TEL # (603) 862-2930 voice/ (603) 862-1527 TTY / 7-1-1 Relay NH, or visit the CREO website. Anonymous reports may be submitted.